Nutanix 3-Node NCI/AOS Cluster Command Notes

Purpose

These notes document the commands visible in the uploaded and identify where each command is run: the Nutanix Controller VM (CVM), the AHV host, or a cluster-wide CLI context.

Important terminology

- NCI is the Nutanix Cloud Infrastructure solution name. The core Nutanix software stack includes AOS, Prism, and a supported hypervisor such as AHV.
- A CVM is the Controller VM. On an AHV node, the CVM provides Nutanix storage and cluster services.
- The AHV host is the hypervisor host. Commands entered in the AHV host shell affect that host only.
- CVM Bash commands normally affect one CVM. `allssh` runs a Bash command on all CVMs. `hostssh` runs a Bash command on all AHV hosts.
- aCLI commands are entered from a CVM and operate on the AHV cluster. Nutanix recommends Prism or supported APIs whenever possible.

Nutanix documents these command locations and the `allssh`/`hostssh` shortcuts in the AHV Command Line Tutorial.

1. Command quick reference

Command or command family	Correct execution location	Purpose	Classification
`cat /etc/nutanix/cluster-version`	CVM Bash, if present	Reads a local Nutanix version file	Read-only diagnostic; not the preferred official version workflow
`cs`	CVM Bash, only if configured as a local alias	Often used as shorthand for cluster status	Do not rely on it; use `cluster status`
`cluster status`	CVM Bash	Shows whether Nutanix cluster	Read-only diagnostic

Command or command family	Correct execution location	Purpose	Classification
		services are up or stopped	
`timedatectl`	CVM or AHV Linux shell	Displays Linux time, timezone, and synchronization state	Read-only Linux diagnostic
`chronyc -n sources -v`	CVM Bash on AOS 6.8+	Verifies Chrony time sources	Read-only time diagnostic
`hostssh "chronyc -n sources -v"`	CVM Bash	Runs the time-source check on every AHV host when Chrony is used	Read-only cluster-wide diagnostic
`ntpq -pn`	CVM/AHV host on older releases	Checks NTP peers before the Chrony transition	Use for releases before AOS 6.8 only
`ip -br a`	CVM or AHV host Linux shell	Compact view of interfaces and IP addresses	Read-only Linux diagnostic
`manage_ovs show_interfaces`	CVM Bash	Displays physical NIC names, link state, and speed for the local AHV host	Read-only AHV networking diagnostic
`allssh "manage_ovs show_interfaces"`	CVM Bash	Displays NIC information from all CVMs/AHV nodes	Read-only cluster-wide diagnostic
`manage_ovs --bridge_name br0 show_uplinks`	CVM Bash	Displays the uplink bond configuration for `br0`	Read-only AHV networking diagnostic
`allssh genesis restart`	CVM Bash	Restarts Genesis on all CVMs	Change operation; use only for a documented maintenance or

Command or command family	Correct execution location	Purpose	Classification
			troubleshooting procedure
`sudo /usr/local/nutanix/cluster/bin/change_cvm_hostname ...`	CVM Bash	Changes the internal CVM hostname	Change operation; CVM reboot is prompted
`change_cvm_display_name ...`	A different CVM Bash session	Changes the CVM name displayed in Prism	Change operation; the target CVM is powered off by the script
`sudo passwd nutanix`	CVM Bash	Changes the CVM `nutanix` account password	Security change; synchronizes to CVMs in the cluster
`ncli cluster get-domain-fault-tolerance-status type=node`	CVM Bash	Checks whether the cluster can tolerate node failure	Read-only health/resiliency check
`ncc health_checks run_all`	CVM Bash	Runs Nutanix Cluster Check health checks	Read-only health check, although some checks can take time
`cluster -s ... --redundancy-factor=2 create`	Legacy/installation context	Attempted cluster-formation command from the notes	Do not use as the current standard deployment procedure without release-specific validation

2. Cluster command locations

CVM Bash shell

SSH to a CVM, for example:

```shell
ssh admin@<CVM_IP>
```

Use the `admin` account for normal administration where supported. Use the `nutanix` account only when a Nutanix procedure specifically requires it.

Examples:

```shell
cluster status
ncli alert ls
ncli cluster get-domain-fault-tolerance-status type=node
ncc health_checks run_all
```

AHV host Bash shell

SSH to an AHV host only for a procedure that explicitly requires host access:

```shell
ssh admin@<AHV_HOST_IP>
```

Commands such as `ip -br a`, `timedatectl`, and host-level `chronyc` are Linux/AHV-host diagnostics. Do not modify AHV networking files or OVS state casually. Nutanix warns that direct AHV host access increases the risk of cluster issues.

aCLI

Start aCLI from a CVM:

```shell
acli
```

Example read-only commands:

```none
host.list
host.get <AHV_HOST_IP>
net.list
```

Exit aCLI with:

```
None
exit
```

`allssh` and `hostssh`

Run these from a CVM Bash shell:

```
Shell
allssh "<command>"
hostssh "<command>"
```

- `allssh` targets all CVMs.
- `hostssh` targets all AHV hosts.
- Prefer them for read-only verification. Do not use a broad fan-out command for a change unless the Nutanix procedure explicitly requires it.

3. Deploying a three-node cluster

What RF2 means

Replication factor 2 maintains two copies of data. Current Nutanix documentation states that RF2 requires at least three nodes for normal node-level fault tolerance. RF3 requires at least five nodes. For a standard three-node cluster, RF2 is normally the appropriate choice unless the deployment design specifies otherwise.

See the Prism Cluster Fault Tolerance guide and Replication Factor guide.

Current recommended deployment method

For a new physical Nutanix cluster, use Nutanix Foundation or Foundation Central. The deployment workflow supplies:

- AOS and AHV images
- Host IP, CVM IP, and IPMI addressing
- Host/CVM VLAN, gateway, and netmask
- CVM timezone
- NTP and DNS servers
- Cluster name
- Cluster virtual IP, if required

- Cluster redundancy/fault-tolerance setting

The Nutanix Foundation cluster deployment guide documents this workflow. The Foundation Central cluster guide also requires at least three selected nodes for cluster creation.

About the handwritten command

The image appears to show a command similar to:

```shell
cluster -s <CVM-IP> --redundancy-factor=2 create
```

Do not copy this command blindly. The current official deployment documents reviewed here describe Foundation/Foundation Central or the Prism cluster-creation workflow, not this exact shell syntax. If a lab guide or an older AOS release specifically provides this command, verify its exact syntax using that release's documentation and `cluster --help` before execution.

A safer documentation statement is:

```none
Create the cluster using Foundation/Foundation Central, select the three nodes,
and select RF2 (or the current equivalent cluster fault-tolerance setting)
according to the installed AOS release.
```

Do not run a cluster-creation command against an existing production cluster. For an existing cluster, use Prism's supported cluster expansion workflow; do not use Foundation to add nodes to a running cluster unless the relevant Foundation procedure explicitly says to do so.

4. Time and NTP commands

Why time synchronization matters

CVMs and AHV hosts require consistent time for cluster communication, logging, failover, authentication, and health checks. Nutanix recommends configuring reliable external NTP sources during deployment and verifying that they are reachable.

Correct verification commands for current releases

On a CVM running AOS 6.8 or later:

```shell
chronyc -n tracking
chronyc -n sources -v
```

Across CVMs:

```shell
allssh "chronyc -n tracking"
allssh "chronyc -n sources -v"
```

On AHV hosts, when Chrony is used:

```shell
hostssh "chronyc -n tracking"
hostssh "chronyc -n sources -v"
```

For older AOS releases before 6.8, the documented check is:

```shell
ntpq -pn
hostssh "ntpq -pn"
```

Nutanix's NCC `check_ntp` guidance states that Chrony replaces NTP in the CVM from AOS 6.8 onward and that `chronyc -n sources -v` should be used instead of `ntpq -pn`.

Commands from the image that should not be used for routine cluster time correction

The image includes commands similar to these:

```shell
timedatectl
hwclock --set --date 'YYYY-MM-DD HH:MM:SS'
hwclock --systohc
date -s 'YYYY-MM-DD HH:MM:SS'
hwclock --hctosys
```

These are generic Linux clock commands, not the preferred Nutanix cluster time-management procedure. They can create time jumps and inconsistent timestamps between CVMs and hosts. Do not manually move a live CVM clock backward or forward to fix an NTP alert. Nutanix specifically warns that a future CVM time can affect in-flight storage metadata operations; contact Nutanix Support for that condition.

Use Prism/Foundation to configure NTP, then use `chronyc` or `ntpq` only for verification according to the AOS version. The Nutanix time-services guidance explains why multiple reliable time sources are important.

5. Checking AHV network interfaces and OVS

Interface summary

From a CVM or AHV Linux shell:

```shell
ip -br a
```

This shows interface names, administrative state, link state, and assigned addresses. It is a Linux diagnostic and does not change configuration.

Check NIC link and speed from a CVM

For the local AHV host:

```shell
manage_ovs show_interfaces
```

For every node:

```shell
allssh "manage_ovs show_interfaces"
```

Typical output includes interface name, mode, link state, and speed. A note such as “eth0 and eth1 have 25K speed” should be interpreted as the actual reported link speed, normally displayed in Mbps or another unit by the command. Verify the real output; do not assume that two interfaces are correctly bonded merely because both show link-up.

Check the `br0` uplink bond

```shell
manage_ovs --bridge_name br0 show_uplinks
allssh "manage_ovs --bridge_name br0 show_uplinks"
```

Nutanix documents `manage_ovs` as an AHV-specific command run from a CVM. The default bridge is normally `br0`, with an uplink bond commonly named `br0-up`. Only use NICs of the same speed in a bond. Nutanix recommends using the Prism Virtual Switch UI for virtual-switch changes on current AOS releases rather than legacy `manage_ovs update_uplinks` workflows.

See OVS Command Line Configuration and AHV Networking Best Practices.

Do not use these for routine configuration

```shell
ovs-vsctl ...
manage_ovs ... update_uplinks
```

`ovs-vsctl` changes the live host configuration and can be lost after a restart. For persistent supported changes, use Prism Virtual Switch or the exact Nutanix procedure for the AOS release. Nutanix also warns against using `allssh` to make broad network changes in production.

6. Genesis and service restart

What Genesis does

Genesis is the Nutanix service manager. Restarting Genesis restarts or reinitializes Nutanix services on the target CVM. It is not a general-purpose “refresh” command and should not be run repeatedly without a reason.

Official syntax is:

```shell
genesis start|stop [all|<service1> [<service2> ...]]|restart|status
```

For a documented cluster-wide procedure:

```shell
allssh "genesis restart"
```

For a single CVM:

```shell
genesis restart
```

Use the exact scope specified by the Nutanix KB or administration guide. Afterward, verify:

```shell
cluster status
ncc health_checks run_all
```

See the AOS Genesis command reference.

7. CVM hostname versus CVM display name

These are different operations.

Change the internal CVM hostname

Run from the CVM being renamed:

```shell
sudo /usr/local/nutanix/cluster/bin/change_cvm_hostname
NTNX-<block-serial>-<position>-CVM
```

Requirements documented by Nutanix:

- The name starts with `NTNX-`.
- The name ends with `-CVM` for a CVM, `-PCVM` for a Prism Central VM, or `-FSVM` for a Files VM.
- The operation prompts for a reboot.
- Verify cluster resiliency before rebooting the CVM.

This changes the hostname inside the CVM. It does not necessarily change the CVM display name shown in Prism.

See How to modify the hostname of a CVM, PCVM, or FSVM.

Change the CVM name displayed in Prism

Run from a different CVM, not from the target CVM:

```
Shell
change_cvm_display_name --cvm_ip=<TARGET_CVM_IP> --cvm_name=NTNX-<new-name>-CVM
```

The script powers off the target CVM while changing the display name. Follow the release-specific post-change instructions for AHV configuration if they apply.

See Changing the Name of the CVM Displayed in Prism.

Correction to the handwritten hostname line

The handwritten command appears to resemble `change_drv_hostname --host-ip ... --host-name ...`. This is not the current documented CVM-hostname command. Do not use it as written. For an AHV host hostname change, use the AHV Administration Guide procedure for the exact AHV release; do not rename the CVM with an AHV-host command.

8. Changing the CVM `nutanix` password

From any CVM:

```
Shell
sudo passwd nutanix
```

This changes the CVM `nutanix` account password and is propagated to the other CVMs in the cluster. Record the new credential securely because it may be required for maintenance, support procedures, and cluster access.

This is not the same as changing:

- The Prism Element `admin` password
- The Prism Central password
- The AHV host `admin`, `root`, or `nutanix` password
- The IPMI/BMC password

For current security guidance, see Modifying Default Passwords and Controller VM Access.

9. Recommended health checks before maintenance

Before restarting a CVM, changing networking, changing a hostname, or stopping the cluster:

```shell
ncli alert ls
ncli cluster get-domain-fault-tolerance-status type=node
ncc health_checks run_all
cluster status
```

The cluster should have no unresolved critical alerts and should be able to tolerate the planned node/CVM operation. Nutanix's AHV cluster health procedure specifically recommends reviewing alerts and verifying node fault tolerance before maintenance.

10. Commands that should not be mixed up

Do not confuse	**Correct distinction**
CVM IP and AHV host IP	The CVM is the Nutanix storage controller VM; the AHV host is the hypervisor.
CVM hostname and CVM Prism display name	Use `change_cvm_hostname` for the internal hostname; use `change_cvm_display_name` for the Prism display name.
CVM `nutanix` password and AHV host password	`sudo passwd nutanix` on a CVM is not the AHV host password procedure.
`allssh` and `hostssh`	`allssh` targets CVMs; `hostssh` targets AHV hosts.
`cluster status` and `cs`	Use the documented full command `cluster status`; `cs` may only be a local alias.
`chronyc` and `ntpq`	Use Chrony checks on AOS 6.8+; use `ntpq` for older releases.
`manage_ovs show_interfaces` and `ip -br a`	`manage_ovs` reports AHV/OVS NIC state; `ip -br a` reports Linux interface/IP state.

Do not confuse	Correct distinction
`cluster stop` and `cluster destroy`	`cluster stop` stops services; `cluster destroy` is destructive and must never be used for routine maintenance.

11. Practical three-node deployment sequence

Use this as a learning sequence, not as a substitute for the release-specific deployment guide:

1. Confirm hardware, AOS, AHV, IPMI, CVM, host IP, VLAN, gateway, DNS, and NTP details.
2. Image and configure the nodes with Foundation or Foundation Central.
3. Select all three nodes for the new cluster.
4. Select RF2 or the current equivalent fault-tolerance setting for a standard three-node design.
5. Configure reliable NTP and DNS sources during deployment.
6. Open Prism after deployment and confirm the cluster is healthy.
7. SSH to a CVM and verify:

```shell
cluster status
ncli cluster get-domain-fault-tolerance-status type=node
chronyc -n tracking
chronyc -n sources -v
allssh "manage_ovs show_interfaces"
allssh "manage_ovs --bridge_name br0 show_uplinks"
ncc health_checks run_all
```

8. Resolve critical alerts and NCC failures before placing a node into maintenance mode or changing cluster networking.
9. Change default credentials using supported procedures and store them in the approved password-management system.

12. Final corrected command list

Safe read-only verification from a CVM

```shell
cluster status
ncli alert ls
```

```
ncli cluster get-domain-fault-tolerance-status type=node
ncc health_checks run_all
chronyc -n tracking
chronyc -n sources -v
allssh "chronyc -n sources -v"
ip -br a
manage_ovs show_interfaces
manage_ovs --bridge_name br0 show_uplinks
allssh "manage_ovs show_interfaces"
allssh "manage_ovs --bridge_name br0 show_uplinks"
```

Supported change examples, only when required by a Nutanix procedure

Shell

```
allssh "genesis restart"
sudo /usr/local/nutanix/cluster/bin/change_cvm_hostname
NTNX-<block-serial>-<position>-CVM
change_cvm_display_name --cvm_ip=<TARGET_CVM_IP> --cvm_name=NTNX-<new-name>-CVM
sudo passwd nutanix
```

Do not use as a routine Nutanix time-fix method

Shell

```
hwclock --set --date 'YYYY-MM-DD HH:MM:SS'
hwclock --systohc
date -s 'YYYY-MM-DD HH:MM:SS'
hwclock --hctosys
```

Official references

- Nutanix AHV Command Line Tutorial
- Creating a Cluster with Foundation
- Foundation Central: Creating Clusters
- Prism Cluster Fault Tolerance
- NCC `check_ntp` Health Check
- OVS Command Line Configuration
- AHV Networking Best Practices
- AOS Genesis Command Reference
- CVM, PCVM, and FSVM Hostname Modification
- CVM Display Name Change
- CVM Access and Security
- AHV Cluster Health Verification

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